# Chit 13 — MongoDB: Teachers, Department, Students

## 1. Create collections and insert sample documents

```
use college_db

// Teachers collection
db.Teachers.insertMany([
  { Tname:'Arun', dno:1, Experience:5, Salary:15000, Date_of_Joining:new Date('2019-06-01') },
  { Tname:'Meena', dno:2, Experience:8, Salary:20000, Date_of_Joining:new Date('2016-01-15') },
  { Tname:'Ravi', dno:2, Experience:3, Salary:12000, Date_of_Joining:new Date('2021-07-10') },
  { Tname:'Sita', dno:1, Experience:10, Salary:25000, Date_of_Joining:new Date('2014-03-20') }
])

// Department collection
db.Department.insertMany([
  { Dno:1, Dname:'Computer' },
  { Dno:2, Dname:'Science' }
])

// Students collection
db.Students.insertMany([
  { Sname:'Amit', Roll_No:1, Class:'FE' },
  { Sname:'xyz', Roll_No:2, Class:'SE' },
  { Sname:'Priya', Roll_No:3, Class:'TE' },
  { Sname:'Riya', Roll_No:4, Class:'BE' },
  { Sname:'Karan', Roll_No:5, Class:'FE' }
])
```

## 2. Find teachers with dno=2 and salary >= 10,000

```
db.Teachers.find({
  dno: 2,
  Salary: { $gte: 10000 }
})
```

## 3. Find student with Roll_No=2 or Sname='xyz'

```
db.Students.find({
  $or: [ { Roll_No: 2 }, { Sname: 'xyz' } ]
})
```

## 4. Update student name where Roll_No=5

```
db.Students.updateOne(
  { Roll_No: 5 },
  { $set: { Sname: 'NewName' } }
)
```

## 5. Delete all students of class 'FE'

```
db.Students.deleteMany({ Class: 'FE' })
```

## 6. Apply index on Students collection

```
db.Students.createIndex({ Roll_No: 1 })
db.Students.getIndexes()
```